

EdTech Study Platform – Full Project Production-Ready Plan

PART 1: WHAT THIS PRODUCT IS

Product name: EdTech Study Platform – Smart-Gated Advanced Exam System

Target users: Indian students aged 16–25 preparing for JEE, NEET, GATE, and similar competitive exams.

Core idea: A Progressive Web App (PWA) that forces mastery-based learning. Students cannot skip ahead – every topic is locked until the previous step is completed and passed.

PART 2: TECH STACK (reconciled across both documents)

Layer	SRS Spec
Frontend Framework	React 18 + Vite
Language	TypeScript
Styling	Tailwind CSS + Framer Motion
State	Zustand
Data Fetching	–
Forms	–
Database	Supabase PostgreSQL + RLS
Auth	Supabase Auth
Backend Logic	Supabase Edge Functions (Deno)

Layer	SRS Spec
File Storage	Backblaze B2
CDN	Cloudflare
Hosting	Vercel
Emails	Resend
Error Monitoring	Sentry
Offline	Dexie.js (IndexedDB) + Service Worker
PDF Rendering	pdf.js + pdf-lib
Testing	–
CI/CD	GitHub Actions

Critical architectural rule (non-negotiable): No Express servers, no Node.js runtimes, no custom JWT, no alternative cloud platforms. Supabase and Backblaze B2 are the **only** backend/storage providers.

PART 3: USER ROLES

Role	Permissions	Status on registration
Super Admin	Full system control; bootstrap owner; user management; role promotion/demotion; view all data; emergency recovery	First registered user only (auto-assigned)
Admin	Create/edit/delete subjects, chapters, topics, exams, questions,	Promoted by Super Admin

Role	Permissions	Status on registration
	solutions, QR codes; configure gates; bulk import; view analytics	
Student	Browse gated content; take exams; upload notes (gated); use Pomodoro; manage study plan; log external tests; manage profile	All subsequent registrations (pending approval)

Key constraint: The very first registration becomes Super Admin automatically via a DB trigger. All subsequent registrations are pending_approval students. No user can self-assign Super Admin via registration.

PART 4: CORE LEARNING FLOW (Gating Logic)

This is the heart of the entire platform. Every step is blocked until the previous one passes:

Subject

└─ Chapter

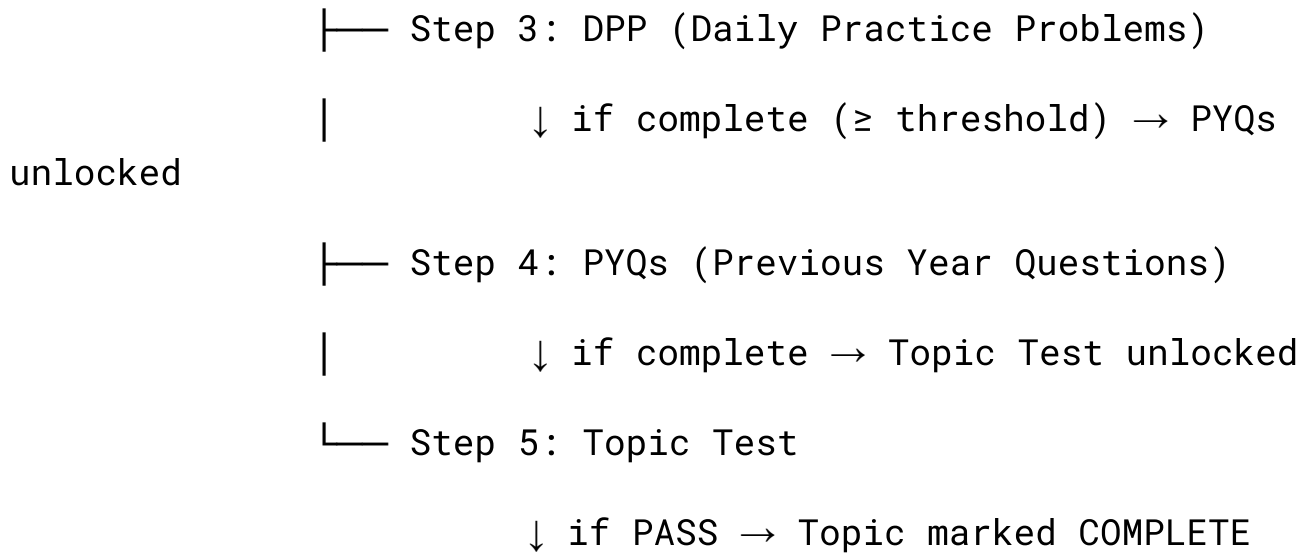
└─ Topic

└─ Step 1: Watch Lecture (Telegram deep link – ALWAYS unlocked)

| ↓ (click logged, but NOT completion)

└─ Step 2: Lecture Quiz (Gate 1 – MUST pass ≥ threshold)

| ↓ if PASS → DPP unlocked



When ALL Topics in a Chapter are complete:

→ Chapter Test UNLOCKED

When Chapter Test is ATTEMPTED:

→ PDF Upload UNLOCKED for that chapter

When ALL Chapters in a Subject are complete:

→ Subject Test UNLOCKED

When ≥ 2 Subjects are complete:

→ Multi-Subject Test UNLOCKED

When ALL Subjects are complete:

→ Grand Test / Full Syllabus Test UNLOCKED

Security: Gate enforcement happens on **both** frontend (UI states) AND backend (Edge Functions return HTTP 403 if prerequisites not met). Frontend lock states are cosmetic; real security is server-side.

5.1 Landing Page & Onboarding

- 5-screen responsive carousel slider (mobile-first)
- Screen 1: Smart Mastery Path
- Screen 2: Real Exam Simulation
- Screen 3: Instant Video Solutions
- Screen 4: Distraction-Free Focus
- Screen 5: CTA (Get Started / Login)
- Progress dots at bottom (5 dots, active = filled)
- Horizontal swipe on mobile; smooth fade/slide-up (300ms ease-out)
- CTA sticky at bottom on mobile; centered on desktop
- Breakpoints: Mobile < 768px | Tablet 768–1024px | Desktop > 1024px

5.2 Registration & Auth

- **Full Name:** required, alphanumeric + spaces, min 2 chars
- **Mobile Number:** required, +91 prefix mandatory, exactly 10 digits after prefix, regex: `^(\\+91)[\\s-]?[6-9]\\d{9}$`, must start with 6-9
- **Email:** required, valid format
- **Password:** min 8 chars, must contain uppercase + lowercase + number + special char, strength meter shown, show/hide toggle
- **Confirm Password:** must match, show/hide toggle
- Register button disabled until all validations pass
- On success: verification email sent via Resend within 30 seconds
- Login blocked until email verified (email_verified = false → 403)
- "Resend Verification Email" button (rate limited: max 3/hour)
- Unverified login shows banner with Resend button
- "Forgot Password?" link → Resend secure reset link

5.3 Profile Management

- **Profile photo:** upload JPG/PNG/WEBP (max 2MB before compression), client-side compress to max 500KB and resize to max 500×500px, stored in Backblaze B2 at `users/{uid}/profile/photo.jpg`
- Upload / Change / Remove (remove reverts to default avatar)

- Only the owner can manage their own photo
- **Display fields:** circular avatar, full name, masked mobile (+91 XXXXX XXXXX), email with verified badge (read-only)
- Name and mobile are editable (with validation); email requires re-verification to change
- **Password Change** (via Profile Menu): current password required (re-auth), new password must meet strength criteria, new password cannot be same as current, show/hide toggles on all 3 fields

5.4 Learning Path Dashboard

- Visual tree of Subject → Chapter → Topic hierarchy
- Three states per item:  Locked (grayed + grayscale, tooltip: "Complete previous step"),  Unlocked (active, clickable, subtle Focus Indigo glow),  Completed (green checkmark)
- Top bar: Focus Streak ( days), Today's Focus time, Profile Avatar
- Today's Plan section: auto-generated + manual tasks, smart suggestions ("Complete Lecture Quiz for Topic X")
- Quick Actions: [Start Pomodoro] (big button), [Add Manual Task]
- Performance Trends: Internal vs. External Test Scores chart

5.5 Exam Interface

- Full-screen, distraction-free environment
- **Header:** Timer (HH:MM:SS countdown), User Name, Pause (max 2 pauses/exam), Submit
- **Main area:** Question text, radio options, "Mark for Review" checkbox
- **Question grid sidebar:** Color-coded per question –  Green (Answered),  Red (Not Answered),  Yellow (Marked for Review),  Gray (Not Visited)
- Direct jump by clicking question number
- **Footer:** Save & Next, Previous, Clear Response
- Mobile: stacked layout, swipe navigation; Desktop: split screen (question left, grid right)

- Keyboard shortcuts: arrow keys to navigate, M to mark, Enter to save
- Timer warnings at 5 mins and 1 min (visual + audible)
- **Auto-submit**: at 00:00, on tab close without pausing, or after 10 mins disconnected
- **Server-client timer sync** to prevent cheating
- Answers saved locally every change (IndexedDB fallback)
- Tab-switch detection with warning (configurable)
- Browser crash recovery: "Resume Exam" prompt on reopen

5.6 Exam Results & Analysis

- **Summary card**: Score, Accuracy, Time, Rank/Percentile
- **Charts**: Pie (Correct/Incorrect/Skipped), Bar (topic-wise breakdown)
- **Tabs**:
 - Answer Sheet: question list with user answer vs. correct answer (color-coded)
 - Solutions: text explanations with LaTeX/MathJax support
 - Videos: QR codes + embedded YouTube players
 - Insights: "Weak in Thermodynamics", "Slow in Calculus"
- Time per question logged and shown
- Negative marking applied correctly
- Filter answer sheet by "Incorrect Only" or "Skipped Only"

5.7 QR Code Video Solutions

- Generated at 4 levels: Subject (overview video), Chapter (summary video), Topic (concept video), Question (specific solution video)
- Admins paste YouTube URL → system auto-generates QR code (SVG + PNG)
- User scans QR → redirects to YouTube
- Fallback: clickable link/button if QR scan fails
- Missing video shows "Video Solution Unavailable" but still shows text solution

5.8 PDF Upload System (Gated)

- Unlocked only after Chapter Test is attempted (pdf_upload_unlocked = true)
- Locked state shows: "Complete Chapter to unlock Notes Upload"
- Files: PDF only, max 10MB per file, max 500MB per user, global max 9GB
- Storage path: users/{uid}/chapters/{chapter_id}/notes_{timestamp}.pdf
- Client-side text extraction via pdf.js for search indexing
- Annotations (highlights/notes) stored as JSON
- File hash (SHA-256) for deduplication
- Presigned B2 URLs expire after 15 minutes
- Deleting a note removes the B2 file
- Cloudflare CDN in front of B2 for caching

5.9 Pomodoro Timer Module

- **Modes:** Standard (25 min work / 5 min break), Long Break (15 min), Custom
- Persists across page refresh (IndexedDB + server sync)
- Works in background via Service Worker
- Audio chime on start/end/break
- Logs to pomodoro_sessions table: user_id, duration_seconds, topic_context, start_time, end_time
- Calculates daily/weekly streaks (min 4 sessions/day = streak)
- "Focus Master" badge for 7-day streak
- Context-aware: if started on a Topic page, session tags that topic
- Dashboard shows "Total Focus Time Today" and "Current Streak"
- Floating widget always accessible; modal with pause/resume/skip break

5.10 Smart Study Planner

- **Auto-generated tasks** (daily cron at 00:00 UTC): fetches weak topics (<50% accuracy in last 3 attempts), fetches spaced repetition due tasks, generates 3–5 tasks, stored with source = 'auto'

- **Manual tasks:** user creates custom tasks (free text, or linked to platform topic/personal topic), stored with source = 'manual'
- Status lifecycle: pending → in_progress → completed → skipped
- Drag & drop reordering (persists)
- Smart suggestions: "You have 2 weak topics in Physics. Want to add DPP for Topic X?"
- **Gating respect:** auto-tasks never suggest locked content; if locked, suggests prerequisite instead ("Complete Lecture Quiz for Topic X" instead of "Attempt DPP")

5.11 Test Tracker

- Logs both internal platform tests and external exam results
- External test log fields: exam_name, exam_date, score_obtained, total_marks, percentile, rank, notes
- Table view: Date | Exam Name | Score | Percentile | Trend (▲/▼)
- Line chart: Internal Test Average vs. External Test Average overlay

5.12 Notification System

- **In-app:** bell icon with unread count
- **Email** (via Resend): Welcome, Verification, Approval, Password Reset, Storage Alert, Resend Verification
- **Web Push:** "Your Daily Plan is ready", "Streak about to break", "Exam starting soon", "Your scheduled test starts in 15 mins"
- Triggers: user approved, new auto-task generated, streak milestone (7 days), B2 storage >8GB (admin only)
- User can toggle: Daily Plan reminders, Streak reminders, Marketing
- System critical alerts (account suspended, email verification required) cannot be disabled
- Delivered within 5 minutes of trigger

5.13 Admin Panel

- CRUD for Subjects, Chapters, Topics

- Telegram deep link configuration per topic (telegram_chat_id + telegram_message_id)
 - Exam/test creation: question pool selection, duration, negative marking, passing score
 - Text solution upload (Markdown/LaTeX), YouTube URL → QR auto-generated
 - Drag-and-drop video ↔ question/topic mapping
 - Bulk CSV/JSON import for questions (must handle 1000+ questions)
 - Answer key bulk upload via CSV
 - Lecture CTR analytics: click-through rates, flag <10% CTR for review
 - Gate configuration (set thresholds, retry limits, difficulty) – no code changes needed, effective immediately
 - User approval queue: Approve / Reject pending users
 - Ban/Suspend users
 - Reset user progress (per topic/chapter/test)
 - View user data (notes, PDFs, exam attempts) for moderation
 - Storage monitor dashboard (B2 usage vs. 9GB limit)
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PART 6: DATABASE SCHEMA (all tables from SRS section 9)

Core Content Tables

- **profiles** – extends auth.users; fields: id, full_name, mobile_number, email, role, is_approved, status, email_verified, created_at, updated_at. Trigger: handle_new_user (first user = super_admin, rest = student pending_approval)
- **subjects** – id, title, description, order_index, icon_url, is_active, created_at/updated_at
- **chapters** – id, subject_id (FK), title, description, order_index, is_active, created_at/updated_at
- **topics** – id, chapter_id (FK), title, description, order_index, is_active, created_at/updated_at

- **lectures** – id, topic_id (FK), telegram_chat_id, telegram_message_id, is_active
- **lecture_clicks** – id, user_id, lecture_id, clicked_at (tracking only)
- **quizzes** – id, topic_id (nullable), chapter_id (nullable), subject_id (nullable), title, type (lecture_quiz|dpp|topic_test|chapter_test|subject_test|grand_test), passing_score, max_attempts, duration_minutes, negative_marking, is_active, created_at
- **quiz_questions** – id, quiz_id (FK cascade), question_text, options (JSONB {"A":"opt1",...}), correct_answer, explanation, difficulty (1–5), order_index, video_solution_url, qr_code_url

Progress & Gating Tables

- **user_topic_progress** – PK: (user_id, topic_id); lecture_clicked, lecture_quiz_passed, lecture_quiz_score, dpp_completed, dpp_score, pyqs_completed, pyqs_score, topic_test_passed, topic_test_score, topic_complete, updated_at
- **user_chapter_progress** – PK: (user_id, chapter_id); all_topics_complete, chapter_test_attempted, chapter_test_passed, pdf_upload_unlocked, updated_at
- **user_subject_progress** – PK: (user_id, subject_id); all_chapters_complete, subject_test_attempted, subject_test_passed, updated_at
- **user_attempts** – id, user_id, quiz_id, content_id, content_type, score, total_marks, accuracy, time_taken_ms, negative_marks_applied, status (in_progress|paused|submitted|expired|abandoned), started_at, submitted_at, is_correct_summary (JSONB)
- **user_answers** – id, attempt_id (FK cascade), question_id, selected_option, is_correct, time_spent_ms, is_marked_for_review

Storage Tables

- **user_notes** – id, user_id, chapter_id, title, content_jsonb, b2_storage_path, b2_file_id, mime_type, pdf_size_bytes,

pdf_text_index, file_hash, tags[], search_vector (generated TSVECTOR for full-text search), created_at/updated_at. Trigger: check_chapter_completion_for_upload blocks insert if pdf_upload_unlocked = false

- **user_profile_photos** – id, user_id (unique), b2_storage_path, b2_file_id, mime_type, size_bytes, created_at/updated_at

Productivity Tables

- **pomodoro_sessions** – user_id, duration_seconds, topic_context (optional), start_time, end_time
- **study_tasks** – id, user_id, title, description, target_type (platform_subtopic|personal_topic|free_text), target_id, status, priority, due_date, source (auto|manual), created_at, completed_at
- **external_tests** – id, user_id, exam_name, exam_date, score_obtained, total_marks, percentile, rank, notes, created_at

System Tables

- **system_config** – key (PK), value (JSONB), updated_at, updated_by; pre-seeded with: uploads_paused, global_storage_limit_gb=9, per_user_storage_limit_mb=500, lecture_quiz_passing_score=60, topic_test_passing_score=70, max_quiz_attempts=3, max_exam_pauses=2, exam_timeout_warning_mins=5, mobile_validation_regex, password_min_length=8, require_email_verification=true
- **audit_logs** – id, actor_id, action, target_type, target_id, old_value (JSONB), new_value (JSONB), ip_address, created_at
- **attempts_archive** – mirror of user_attempts, records >12 months old archived monthly

PART 7: SUPABASE EDGE FUNCTIONS (all 13 required)

Function	Purpose
on-user-created	Super Admin bootstrap + mobile number validation
send-verification-email	Custom email verification logic via Resend
b2-upload	Quota check + gate access validation + B2 signed URL generation
b2-profile-photo	Image compression + upload to B2
b2-download	Presigned URL for PDFs and profile photos
storage-monitor	Daily 10GB threshold alert to admin
gate-checker	Real-time prerequisite validation (must respond <50ms)
exam-engine	Timer sync, auto-submit, state management
qr-generator	Dynamic QR code mapping for solutions
telegram-track	Log click events (NOT completion)
study-planner	Daily cron: auto-generate study tasks
pomodoro-sync	Session logging
rate-limiter	Abuse prevention

PART 8: API ENDPOINTS (all from section 7.4.3)

Method	Endpoint	Purpose
POST	/api/gate/check	Validate prerequisites

Method	Endpoint	Purpose
POST	/api/exam/start	Start exam (returns token + timestamp)
POST	/api/exam/submit	Submit answers
GET	/api/exam/results/{result_id}	Fetch results
POST	/api/qr/generate	Generate QR code
POST	/api/b2/upload-url	Get presigned URL for PDF upload
POST	/api/b2/download-url	Get presigned URL for PDF download
POST	/api/b2/profile-upload-url	Get presigned URL for photo upload
POST	/api/b2/profile-download-url	Get presigned URL for photo download
GET	/api/b2/quota	Check storage quota
POST	/api/auth/verify-email	Email verification

PART 9: ENVIRONMENT VARIABLES REQUIRED

Variable	Used By
SUPABASE_URL	Client + Edge Functions
SUPABASE_ANON_KEY	Client only
SUPABASE_SERVICE_ROLE_KEY	Edge Functions only (never client)

Variable	Used By
B2_ACCOUNT_ID	Edge Functions
B2_APPLICATION_KEY_ID	Edge Functions
B2_APPLICATION_KEY	Edge Functions (never client)
B2_BUCKET_NAME	Edge Functions
RESEND_API_KEY	Edge Functions
SENTRY_DSN	Client + server

PART 10: NON-FUNCTIONAL REQUIREMENTS

Requirement	Target
Initial users	20 concurrent
Scale target	1,000+ users
Availability	99.9% uptime during exam windows
Gate check latency	< 50ms
Page load (3G)	< 2 seconds
Exam interface load	< 1 second
API response (p95)	< 200ms
Email verification send	< 30 seconds
Notification delivery	< 5 minutes

Requirement	Target
PDF text extraction	< 30 seconds
Study planner generation	< 5 minutes
Quiz retry limit	Max 5 attempts/minute/user
PDF uploads	Max 5/hour/user
Registration	Max 10/IP/hour
Password change	Max 3/hour
Profile photo upload	Max 5/hour
B2 storage (global)	≤ 10GB (alert at 9.5GB)
B2 storage (per user)	≤ 700MB hard limit
Max file size per PDF	30MB
Signed URL expiry	15 minutes
Infrastructure cost	\$0/month (free tiers)
Test code coverage	> 90% for Edge Functions + triggers
Accessibility	WCAG 2.1 AA
Browser support	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Mobile OS	iOS 14+, Android 10+

PART 11: UI DESIGN SYSTEM (exact spec)

Default theme: Dark Mode (system-preference aware, CSS variables for switching)

Role	Color Name	Hex
Background	Deep Slate	#0F172A
Surface/Card	Midnight Blue	#1E293B
Surface/Light	Soft Gray	#F1F5F9
Primary	Focus Indigo	#6366F1
Primary Hover	Deep Indigo	#4F46E5
Secondary	Calm Teal	#14B8A6
Accent	Soft Purple	#A855F7
Text Primary	Off-White	#F8FAFC
Text Secondary	Slate Gray	#94A3B8
Text Disabled	Dim Gray	#475569
Border	Muted Border	#334155
Success	Growth Green	#10B981
Warning	Caution Amber	#F59E0B
Error	Alert Red	#EF4444

PART 12: IMPLEMENTATION PLAN – 9 PHASES

1. **Phase 1 – SRS Analysis:** Deep extraction of all requirements (already done)

2. **Phase 2 – System Architecture:** Full serverless architecture (Supabase-first, Edge Functions minimal)
3. **Phase 3 – SQL Backend:** One complete production-ready SQL schema with extensions, enums, tables, triggers, functions, views, RLS policies, seed data, permissions
4. **Phase 4 – Implementation Roadmap:** Module-by-module: Project Setup → Auth → Onboarding → Dashboard → Main Modules → File Uploads → Notifications → Realtime → Analytics → Admin Panel → Optimization → Security Hardening → Testing → Deployment
5. **Phase 5 – AI Prompt Library:** Copy-paste prompts for each module with exact file names, folder structure, TypeScript types, validation rules, edge cases
6. **Phase 6 – GitHub Collaboration:** Monorepo structure, branch strategy, commit conventions, PR workflow, issue templates, code review checklist
7. **Phase 7 – Testing Strategy:** Unit (>90% coverage), Integration (E2E flows), Load (k6, 2000 concurrent), Security (OWASP, pen test), Usability (WCAG 2.1 AA)
8. **Phase 8 – DevOps & Deployment:** Vercel (frontend), Supabase Cloud (DB + Auth + Edge Functions), Backblaze B2 (storage), GitHub Actions (CI/CD), Sentry, Pingdom
9. **Phase 9 – Documentation:** Developer docs, API docs, SQL docs, deployment guide, environment setup guide, README

The SRS is the authoritative product specification – it was written first and approved for production. The implementation plan was generated as an execution blueprint and chose Next.js (for its App Router, SSR, and Vercel-native deployment benefits), which is a reasonable upgrade but deviates from what the SRS specifies.

Everything else is consistent. Both documents agree on: Supabase, Backblaze B2, Tailwind CSS, Zustand, Cloudflare CDN, Vercel, Resend, Sentry, GitHub Actions, and the full serverless constraint.

PART 13: WHAT WOULD YOU LIKE TO BUILD FIRST?